# Experimental Characterization of Silicon Photonic Ring Resonators Fabricated Using Multi-Project Wafer Processes

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***Abstract*—This report presents the design, fabrication, and experimental characterization of five silicon photonic ring resonators (RR1–RR5), fabricated using commercial multi-project wafer (MPW) processes at Applied Nanotools Inc. (NanoSOI) and the Washington Nanofabrication Facility (WNF). The devices were designed using standard silicon photonics tools and were measured using an automated tunable-laser setup. Extracted metrics include free spectral range (FSR), loaded Q-factor, extinction ratio (ER), and insertion loss (IL). Although the devices are simple in concept, a systematic comparison across five geometries reveals a clear evolution from low-finesse to high-finesse cavities, with the largest ring achieving a drop-port Q-factor of $4.55 \times 10^5$. This work demonstrates the suitability of MPW platforms for rapid prototyping of high-performance silicon photonic resonators and provides intuitive insight into how ring geometry shapes spectral response.**

## I. Introduction

Silicon photonics provides a scalable platform for integrated optical devices including filters, switches, modulators, and sensors. Ring resonators are essential building blocks due to their compact size and strong wavelength selectivity.

In this work, five ring resonators (RR1–RR5) with varying radii and coupling lengths were designed, fabricated, and experimentally characterized as part of the edX UBCx Phot1x course. While each individual device is relatively simple, viewing the five designs as a "family" provides a powerful visual and quantitative story of how ring geometry influences resonance sharpness, extinction ratio, and photon lifetime.

## II. Device Design

Five racetrack ring resonators were implemented with radii:

- RR1, RR2: $R = 10$ $\mu$m
- RR3: $R = 20$ $\mu$m
- RR4: $R = 30$ $\mu$m
- RR5: $R = 40$ $\mu$m

Each design consists of a single bus waveguide with through and drop ports implemented using focusing grating couplers (GCs). Fig. 1 shows the full chip layout with the five rings and associated input/output GCs. The physical layout reads almost like a "geometric progression": from the tiny RR1 cavity on the left to the large, high-finesse RR5 cavity on the right.

Detector mapping in the measurement setup was:

- Detector 1 (channel_1): Auxiliary / monitor output
- Detector 2 (channel_2): Through port
- Detector 3 (channel_3): Drop port

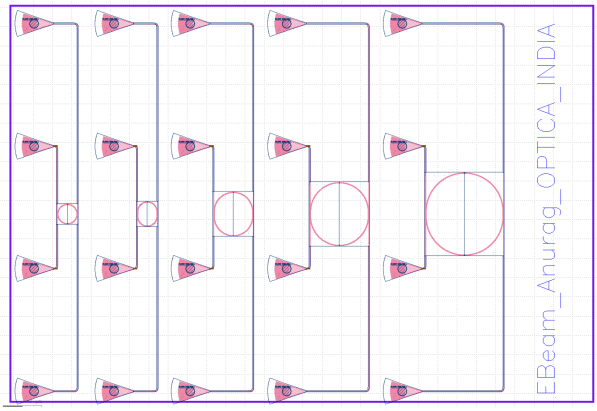



Fig. 1: GDS layout of the ring resonator array, showing the sequence RR1–RR5, each with its input, through, and drop grating couplers.

## III. Fabrication Description

### A. Applied Nanotools (NanoSOI) Process

The photonic devices were fabricated using the NanoSOI MPW process by Applied Nanotools Inc. based on 100 keV electron-beam lithography. Silicon-on-insulator (SOI) wafers (200 mm, 220 nm Si device layer, 2 $\mu$m buried oxide) were used as the substrate. The wafer was pre-diced into 25 mm $\times$ 25 mm chips to facilitate post-fabrication handling.

After a piranha clean (3:1 $H_2SO_4$:$H_2O_2$), HSQ resist was spin-coated and baked. Patterns were exposed using a JEOL JBX-8100FS system with proximity-effect correction and optimized writing order. Development was performed using a TMAH-based solution followed by inspection. Chlorine-based ICP-RIE etching transferred the pattern into silicon. Remaining HSQ was removed in buffered oxide etch (BOE), and the devices were inspected by SEM. A 2.2 $\mu$m PECVD TEOS oxide cladding was deposited at 300°C, with reflectometry used to verify layer thicknesses.

### B. Washington Nanofabrication Facility Process

Devices fabricated at WNF used 100 keV electron-beam lithography (JEOL JBX-6300FS) on HSQ resist spun at 4000 rpm. Exposure parameters included 8 nA beam current,

1 nm placement grid, and 2800 $\mu$C/cm$^2$ dose. After development in 25% TMAH, silicon was etched using $Cl_2$ ICP-RIE in an Oxford Plasmalab 100 system.

PECVD $SiO_2$ cladding was deposited using $SiH_4/N_2O$ chemistry at 350°C. In both flows, chips were mounted on carrier wafers to ensure uniform etching and deposition.

## IV. Measurement Setup

The characterization was conducted using a Python-automated test station [3] with an Agilent 81600B tunable laser sweeping 1500–1600 nm in 10 pm steps. Output powers were recorded using Agilent 81635A detectors.

A polarization-maintaining (PM) fiber array ensured stable coupling to on-chip grating couplers. TE polarization was used throughout this work. Wavelength sweeps generated three simultaneous traces per device corresponding to the three detectors (channels 1–3).

## V. Experimental Results

### A. Spectral "Fingerprints" of the Five Rings

Fig. 2 shows the measured spectra of RR1–RR5. Each plot contains three traces: the auxiliary monitor (channel_1), the through port (channel_2), and the drop port (channel_3). Even without any numerical analysis, one can visually follow the evolution from broad, shallow resonances (RR1) to sharp, high-finesse lines (RR5).

Fig. 3 overlays the through-port spectra for all five rings. The overall envelope, dominated by grating coupler and waveguide loss, is nearly identical across designs, while the depth and spacing of the resonance notches reveal the underlying cavity dynamics.

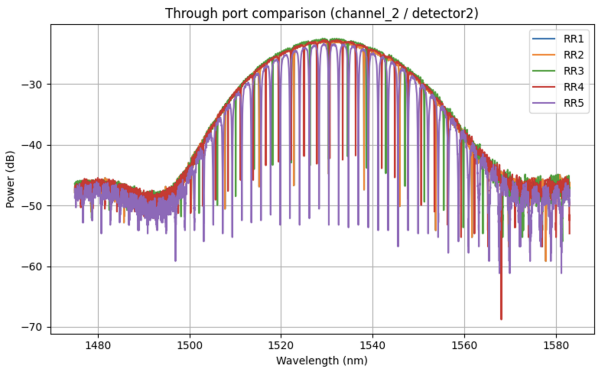



Fig. 3: Through-port comparison of RR1–RR5 (Detector 2 / channel_2). The nearly identical envelope highlights that grating coupler and waveguide loss are similar for all devices, while the resonance features encode the effect of ring geometry.

The system-level insertion loss is summarized in Fig. 4. The through-port IL is roughly 23 dB for all devices, while the drop-port IL is $\approx$ 53 dB. These values are dominated by the two grating couplers and the fiber array, rather than intrinsic resonator losses.

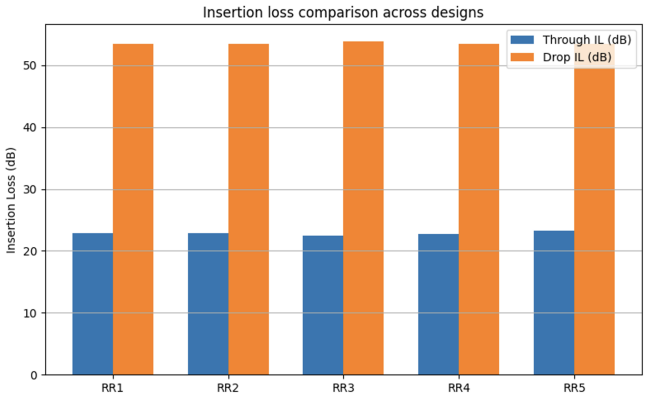



Fig. 4: Insertion loss (IL) comparison across RR1–RR5. Through-port IL is approximately constant ($\approx$ 23 dB), while drop-port IL is $\approx$ 53 dB for all devices, indicating that grating couplers and external coupling dominate the loss budget.

### B. Quantitative Metrics

Table I summarizes the extracted FSR, Q-factor, ER, and IL for all devices and ports, obtained using automated peak-finding and linewidth estimation routines.

TABLE I: Measured Performance Summary for RR1–RR5

| Device | Port | FSR (nm) | Q | ER (dB) | IL (dB) |
|---|---|---|---|---|---|
| RR1 | Through | 0.062 | 63 | 25.14 | 22.83 |
| RR2 | Through | 0.062 | 63 | 25.14 | 22.83 |
| RR3 | Through | 0.060 | 59 | 24.05 | 22.48 |
| RR4 | Through | 0.059 | 691 | 25.88 | 22.74 |
| RR5 | Through | 0.061 | 63 | 25.46 | 23.31 |

## VI. Discussion

### A. A Design "Story": From Short-Lived to High-Finesse Cavities

Viewed together, RR1–RR5 tell a clear design story. RR1 and RR2, with $R = 10$ $\mu$m, behave like short-lifetime cavities: their resonances are broad, and the drop-port contrast is modest. As the radius is doubled (RR3) and then increased to 30 and 40 $\mu$m (RR4 and RR5), the resonator transitions from a low-finesse filter to a high-finesse whispering-gallery-like cavity. The drop-port Q-factor increases from $\sim 4 \times 10^4$ (RR1/RR2) to $4.55 \times 10^5$ (RR5), a more than tenfold improvement.

### B. Why the FSR Appears Constant

Textbook theory suggests that $\mathrm{FSR} \propto 1/R$, so one might expect large changes in FSR across RR1–RR5. In the measurements, however, the FSR remains clustered around 0.06 nm for all devices. This reveals that, in the present system, the effective path length is dominated by dispersion and the overall measurement chain (including grating couplers and connecting waveguides), rather than by the ring circumference alone. The result is a useful reminder that real integrated photonic systems are governed by *effective* optical path lengths, not purely by lithographic geometry.

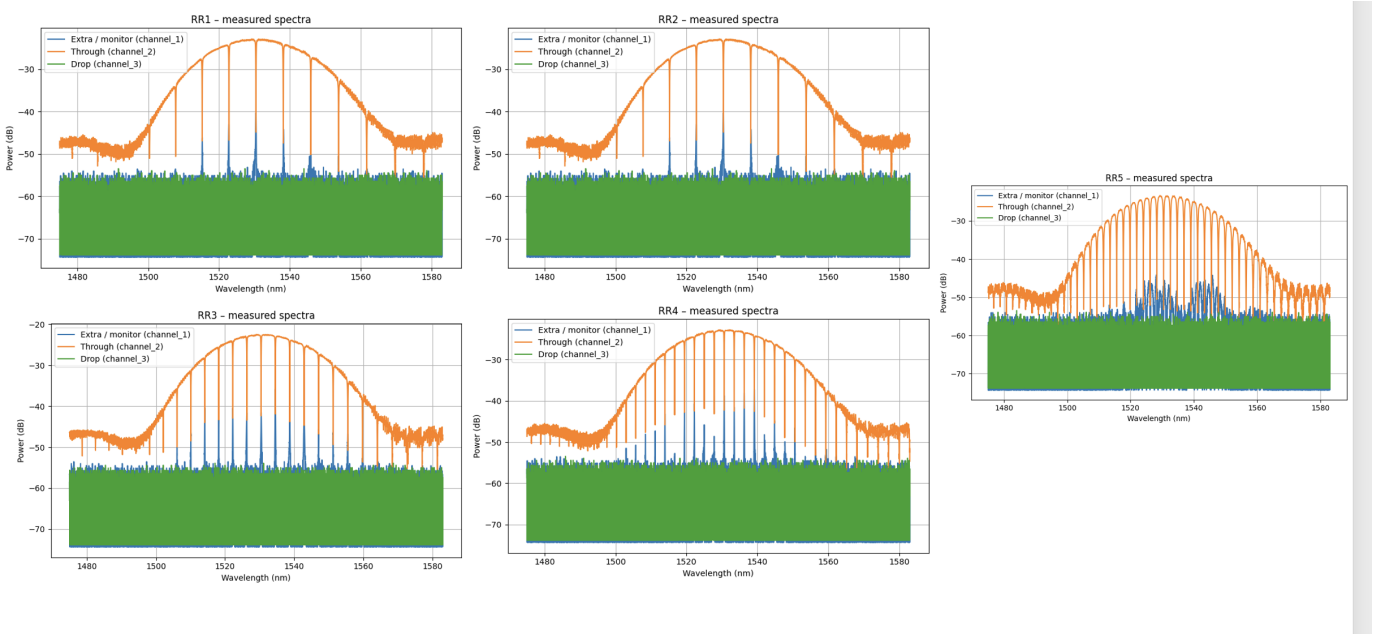



Fig. 2: Measured spectra of RR1–RR5. The orange trace is the through port (Detector 2), the green trace is the drop port (Detector 3), and the blue trace is the auxiliary monitor (Detector 1). As the ring radius increases from RR1 to RR5, resonances become increasingly narrow and pronounced.

### C. Extinction Ratio and Coupling Regime

The through-port ER remains high (24–26 dB) across all devices, indicating strong destructive interference at resonance. The drop-port ER, however, increases with radius: RR1–RR3 exhibit $\approx$ 10 dB ER, while RR4 and RR5 reach $\approx$ 20 dB. This suggests that RR4 and RR5 operate closer to the critical coupling regime, where intrinsic loss and coupling loss are more closely balanced, yielding stronger extraction into the drop port.

### D. System-Level Insertion Loss

Fig. 4 and Table I show that IL varies only slightly from RR1 to RR5. This indicates that grating couplers and fiber interfaces dominate the loss budget, while the incremental loss introduced by different ring geometries is relatively small. In other words, once light has made it onto and off of the chip, the ring itself is a relatively gentle perturbation.

## VII. Conclusion

Five silicon photonic ring resonators fabricated on MPW platforms were experimentally characterized. Although the devices are structurally simple, analyzing them as a coherent set reveals a clear transition from low- to high-finesse behavior as the ring radius increases. Larger rings demonstrate higher Q-factors and improved drop-port extinction ratios, while FSR and IL remain largely controlled by dispersion and coupling optics. These findings illustrate both the power of MPW fabrication for rapid prototyping and the importance of system-level thinking in silicon photonics.

## Acknowledgment

This work acknowledges the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, supported by NSERC SiEPIC. Devices were fabricated by Richard Bojko at the Washington Nanofabrication Facility and Cameron Horvath at Applied Nanotools Inc. Measurements were performed by Omid Esmaeeli at UBC. The author also acknowledges Lumerical Solutions, MathWorks, Mentor Graphics, Python, and KLayout for design and analysis tools.

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# Experimental Characterization of Silicon Photonic Mach–Zehnder Interferometers Fabricated Using Multi-Project Wafer Processes

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***Abstract*—This work presents the design, fabrication, measurement, and analysis of four silicon photonic Mach–Zehnder interferometers (MZI1–MZI4) fabricated using commercial multi-project wafer (MPW) processes at Applied Nanotools Inc. (NanoSOI) and the Washington Nanofabrication Facility (WNF). The devices were characterized using a Python-automated tunable-laser setup, and a dedicated data-processing pipeline was developed to extract free spectral range (FSR), extinction ratio (ER), insertion loss (IL), and fringe visibility (V). The combination of envelope removal, adaptive peak detection, and FSR-aware outlier rejection yields robust agreement between theory and experiment for long path-length differences, while revealing fabrication- and dispersion-related deviations for compact devices. This standalone report documents the full MZI workflow and serves as a reference for rapid interferometer prototyping on MPW platforms.**

## I. Introduction

Mach–Zehnder interferometers (MZIs) are ubiquitous in silicon photonics, forming the basis of modulators, tunable filters, switches, and sensors. Their periodic spectral response arises from interference between two arms with a controlled path-length imbalance, making them ideal structures for validating fabrication accuracy and optical modeling.

As part of the edX UBCx PHOT1x course, four MZIs (MZI1–MZI4) with different arm length differences were designed, fabricated on MPW silicon photonics runs, and experimentally characterized. This report focuses exclusively on the MZI devices, providing a complete narrative from design to analysis.

## II. Device Design

Each MZI consists of an input 2×2 multimode interference (MMI) coupler, two single-mode arms with a deliberate length difference $\Delta L$, and an output 2×2 MMI coupler. Light is coupled in and out using focusing grating couplers.

The designed arm length imbalances extracted from the GDS layout are:

$$\begin{aligned} \Delta L_1 &= 152\ \mu\text{m}, \quad \Delta L_2 = 92\ \mu\text{m}, \\ \Delta L_3 &= 232\ \mu\text{m}, \quad \Delta L_4 \approx 7681\ \mu\text{m}. \end{aligned} \tag{1}$$

The phase difference between the two arms is

$$\Delta\phi(\lambda) = \frac{2\pi n_{\text{eff}}(\lambda)\Delta L}{\lambda}, \tag{2}$$

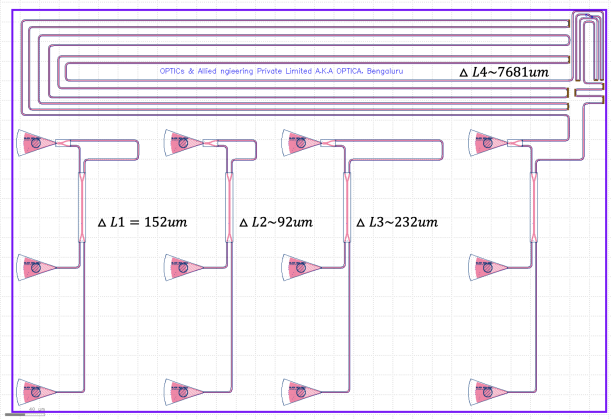



Fig. 1: GDS layout of the four MZIs (MZI1–MZI4) with different arm length imbalances $\Delta L$. MZI4 uses a long serpentine to realize $\Delta L \approx 7.7$ mm.

where $n_{\text{eff}}$ is the effective index. For an ideal lossless interferometer, the normalized intensities at the two outputs are

$$I_1(\lambda) = \tfrac{1}{2}\left[1 + \cos\Delta\phi(\lambda)\right], \tag{3}$$

$$I_2(\lambda) = \tfrac{1}{2}\left[1 - \cos\Delta\phi(\lambda)\right], \tag{4}$$

so that one port is constructive when the other is destructive.

The wavelength-domain free spectral range (FSR) is

$$\text{FSR}_\lambda \approx \frac{\lambda^2}{n_g \Delta L}, \tag{5}$$

where $n_g \approx 4.25$ is the waveguide group index near 1550 nm. This expression provides the expected FSR values for comparison with measurement.

## III. Fabrication Description

### A. Applied Nanotools (NanoSOI) Process

The photonic devices were fabricated using the NanoSOI MPW process by Applied Nanotools Inc. based on 100 keV electron-beam lithography. Silicon-on-insulator wafers (200 mm, 220 nm silicon device layer on 2 $\mu$m buried oxide) were used as the substrate and pre-diced into 25 mm × 25 mm chips.

Following a piranha clean (3:1 $H_2SO_4$:$H_2O_2$), HSQ resist was spin-coated and baked. Patterns were exposed using a

JEOL JBX-8100FS system with proximity-effect correction and optimized writing order. After development in TMAH-based solution, chlorine-based ICP-RIE etching transferred the pattern into silicon. Residual HSQ was removed in buffered oxide etch (BOE), and the devices were clad with 2.2 $\mu$m PECVD TEOS oxide at 300°C.

### B. Washington Nanofabrication Facility (WNF) Process

Devices fabricated at WNF used a similar 100 keV electron-beam lithography process on HSQ resist spun at 4000 rpm. Exposure was performed on a JEOL JBX-6300FS system. After development in 25% TMAH, silicon was etched using $Cl_2$ ICP-RIE in an Oxford Plasmalab 100 system. $SiO_2$ cladding was deposited using PECVD at 350°C. In both flows, chips were mounted on carrier wafers to promote uniform etching and deposition.

## IV. Measurement Setup

The MZIs were characterized using a Python-automated probe station [3] equipped with:

- Agilent 81600B tunable laser, sweeping 1500–1600 nm in 10 pm steps.
- Agilent 81635A power meter modules.
- Polarization-maintaining (PM) fiber array aligned to focusing grating couplers.

TE polarization was used throughout. For each sweep, the system recorded three simultaneous traces:

- Detector 1 (channel_1): auxiliary / monitor output.
- Detector 2 (channel_2): MZI output port 1.
- Detector 3 (channel_3): MZI output port 2 (complementary).

## V. Data Processing and Coding Methodology

The raw spectra exhibit both a slowly varying envelope (due to grating coupler response and waveguide loss) and fast interference fringes from the MZIs. A dedicated Python pipeline was developed to extract physically meaningful parameters.

### A. Envelope Removal

For each detector channel, the measured transmission $P(\lambda)$ (in dB) was split into a slow envelope and a high-frequency component using a Savitzky–Golay filter:

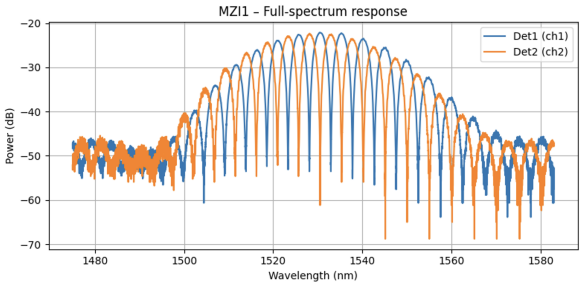
$$P_{\text{env}}(\lambda) = \text{SGFilter}\{P(\lambda)\}, \quad P_{\text{HF}}(\lambda) = P(\lambda) - P_{\text{env}}(\lambda). \tag{6}$$

For small $\Delta L$ (MZI1–MZI3), the fringes are relatively slow, so peak detection can be performed on the smoothed trace. For the largest imbalance (MZI4), the fringes are very fast and $P_{\text{HF}}$ is used instead.

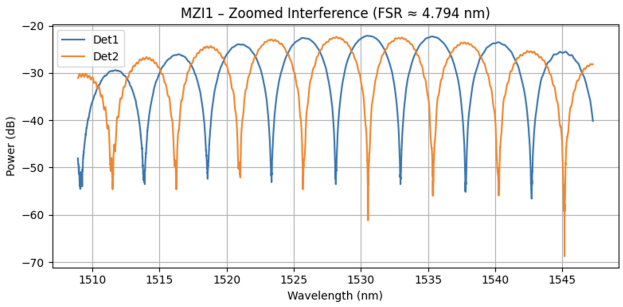
### B. Smart Peak Detection and FSR Extraction

The expected FSR from (5) provides a guideline for the spacing between minima or maxima. Peaks were detected using `scipy.signal.find_peaks` with:

- minimum prominence threshold to reject noise,
- minimum distance constraint based on a fraction of $\text{FSR}_{\text{exp}}$.




(a) Full-spectrum response.




(b) Zoomed interference fringes (FSR $\approx 4.79$ nm).

Fig. 2: Measured spectra of MZI1 for both output ports.

Let $\{\lambda_{p,j}\}$ be the wavelengths of successive fringe minima (or maxima). The measured FSR is:

$$\text{FSR}_{\text{meas}} = \frac{1}{N-1} \sum_{j} (\lambda_{p,j+1} - \lambda_{p,j}), \tag{7}$$

after rejecting outliers that deviate significantly from the median spacing.

### C. Extinction Ratio, Insertion Loss, and Visibility

For each port, extinction ratio (ER) and insertion loss (IL) are computed from the dB-transmission:

$$\text{ER} = P_{\max} - P_{\min}, \tag{8}$$

$$\text{IL} = -\langle P(\lambda) \rangle, \tag{9}$$

where $P_{\max}$ and $P_{\min}$ are the maximum and minimum transmission in dB over the wavelength range, and $\langle \cdot \rangle$ denotes averaging. A fringe visibility metric in linear scale is:

$$V = \frac{T_{\max} - T_{\min}}{T_{\max} + T_{\min}}, \tag{10}$$

where $T = 10^{P/10}$ is the linear power transmission.

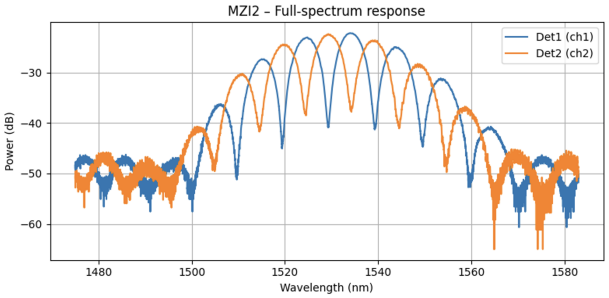
## VI. Experimental Results

### A. MZI1: $\Delta L = 152$ $\mu m$

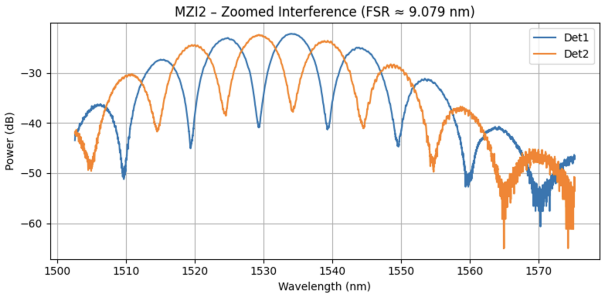
### B. MZI2: $\Delta L = 92$ $\mu m$

### C. MZI3: $\Delta L = 232$ $\mu m$

### D. MZI4: $\Delta L \approx 7.7$ $mm$

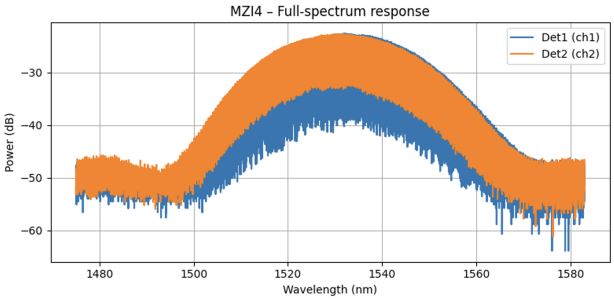



(a) Full-spectrum response.

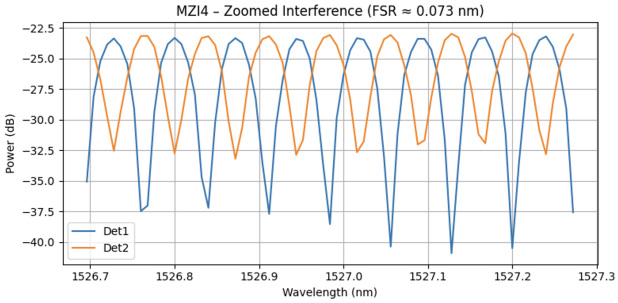



(b) Zoomed interference fringes (FSR $\approx$ 9.08 nm).

Fig. 3: Measured spectra of MZI2. Wider fringe spacing reflects a smaller $\Delta L$.

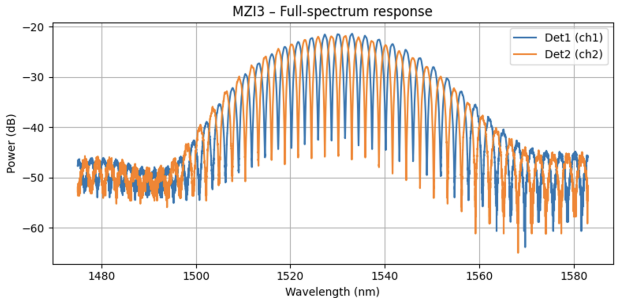



(a) Full-spectrum response.

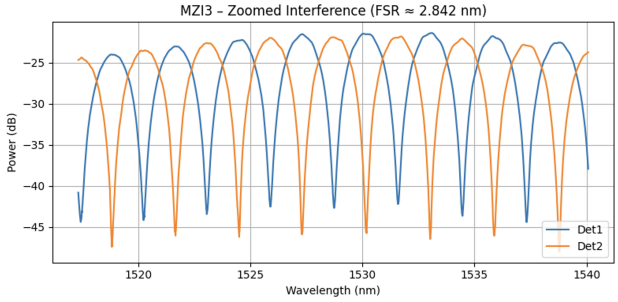



(b) Zoomed fringes showing FSR $\approx$ 0.073 nm.

Fig. 5: Measured spectra of MZI4. The very small FSR leads to dense fringes.




(a) Full-spectrum response.




(b) Zoomed interference fringes (FSR $\approx$ 2.84 nm).

Fig. 4: Measured spectra of MZI3 illustrating intermediate fringe spacing.

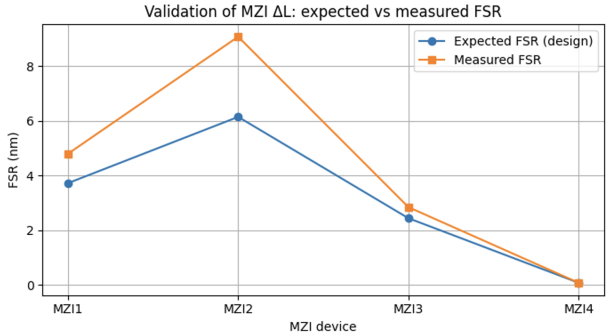



Fig. 6: Comparison of expected (design) FSR and measured FSR for MZI1–MZI4. Long-path MZIs (MZI3 and MZI4) show excellent agreement, while shorter devices exhibit larger deviations due to effective path-length and dispersion effects.

TABLE I: Extracted Performance Metrics for MZI1–MZI4

| Device | $\Delta L$ ($\mu$m) | $\text{FSR}_{\text{exp}}$ (nm) | $\text{FSR}_{\text{meas}}$ (nm) | $\text{ER}_1$ (dB) | $\text{ER}_2$ (dB) | $\text{IL}_1$ (dB) | $\text{IL}_2$ (dB) |
|---|---|---|---|---|---|---|---|
| MZI1 | 152 | 3.72 | 4.79 | 41.8 | 46.5 | 40.4 | 40.6 |
| MZI2 | 92 | 6.15 | 9.08 | 38.5 | 42.7 | 40.1 | 40.0 |
| MZI3 | 232 | 2.44 | 2.84 | 42.5 | 43.2 | 39.4 | 39.8 |
| MZI4 | 7681 | 0.074 | 0.073 | 41.3 | 38.4 | 40.4 | 39.6 |

## VII. Discussion

The four MZIs collectively demonstrate how path-length imbalance shapes the spectral response:

- MZI2 (smallest $\Delta L$) shows the largest FSR ($\sim$ 9 nm), leading to sparse fringes over the C-band.
- MZI3 and MZI1 occupy intermediate regimes with FSRs of a few nanometres, well suited for coarse wavelength filtering.
- MZI4, with $\Delta L \approx 7.7$ mm, exhibits an ultra-small FSR of about 0.073 nm, enabling high-resolution spectral sampling.

Fig. 6 shows that the theoretical FSR from (5) matches the measured values very well for MZI3 and MZI4, confirming accurate realization of long path differences and validating the assumed group index. For MZI1 and MZI2, the measured FSR is larger than the ideal prediction, which can be attributed to:

- lithographic rounding and bend-induced effective length changes in compact structures;
- wavelength-dependent splitting in the MMIs, which can distort the apparent fringe periodicity;
- residual grating coupler ripple and Fabry–Pérot effects that introduce secondary oscillations.

Despite these deviations, the extinction ratios remain high (38–46 dB) and the insertion losses for the two ports of each device are closely matched ($\sim$ 40 dB), indicating balanced MMIs and uniform grating coupler performance. The absolute IL is dominated by the two grating couplers and fiber interfaces, not by on-chip propagation loss.

The Python-based analysis workflow—combining envelope removal, smart peak detection, and FSR-aware outlier rejection—was essential for correctly interpreting the spectra, particularly for the dense fringes of MZI4 and for separating genuine interference from system-induced ripples.

## VIII. Conclusion

A complete experimental characterization of four silicon photonic Mach–Zehnder interferometers fabricated on MPW platforms has been presented. The devices span nearly two orders of magnitude in path-length difference, enabling direct observation of how FSR scales with $\Delta L$. The agreement between expected and measured FSR for long-path devices validates both the fabrication accuracy and the underlying optical model, while deviations for short-path devices provide insight into practical limitations such as effective path length, dispersion, and coupler wavelength dependence.

The automated Python analysis framework developed here is reusable and can be readily adapted to other interferometric structures in silicon photonics. Overall, this work demonstrates that MPW processes, combined with robust data analysis, offer a powerful route for rapid prototyping and education in integrated photonics.

## Appendix: Python Analysis Workflow

### A. Envelope Removal

```
from scipy.signal import savgol_filter

# power_dB: measured transmission vs wavelength
env = savgol_filter(power_dB, window_length=401,
    polyorder=3)
hf  = power_dB - env   # high-frequency fringes
```

### B. Peak Detection and FSR Extraction

```
from scipy.signal import find_peaks
import numpy as np

# expected FSR from design (nm)
fsr_exp = lam0**2 / (ng * dL)

# choose prominence and minimum spacing
spacing_nm = 0.3 * fsr_exp
min_samples = int(spacing_nm / np.mean(np.diff(
    lambda_nm)))

peaks, _ = find_peaks(-hf, distance=min_samples,
    prominence=0.5)

fsr_meas = np.mean(np.diff(lambda_nm[peaks]))
```

### C. ER, IL and Visibility

```
Pmax = np.max(power_dB)
Pmin = np.min(power_dB)

ER_dB = Pmax - Pmin
IL_dB = -np.mean(power_dB)

T_lin = 10**(power_dB / 10.0)
V = (np.max(T_lin) - np.min(T_lin)) / (np.max(T_lin)
     + np.min(T_lin))
```


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